

Anchoring Options to an Incomplete Forward Curve: Identification and Out-of-Sample Evidence from Turkish Electricity

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Abstract

Electricity options in emerging markets have to be valued without a traded option surface and with a forward curve observed only as a strip of monthly averages. A framework is developed that separates what such a market identifies from what it does not. An hourly curve is built as the smoothest one that reproduces the Turkish monthly baseload quotes exactly, and a two-regime Ornstein–Uhlenbeck residual with transition probabilities driven by residual demand is attached in centred form, so that the expected price equals the forward for every parameter vector and under changes of measure in the drift or the switching intensities. Single-hour options are priced through a coupled moment system and finite differences. Independent re-estimation shows that the inherited mean-reversion speed and volatility are individually misidentified but reproduce the stationary dispersion observed in 2025, so values at 24 to 72 hours are insensitive to the former. Out of sample in 2026, the forward curve exceeds realised prices by 385 to 794 TRY/MWh on average and carries all of the bias, while the dispersion assumed proportional to the price level is too narrow by up to a factor of 1.55. The unquoted front month bounds the precision of any reported value.

Keywords: electricity options, forward curve, regime switching, time-varying transition probabilities, out-of-sample evaluation, Turkish electricity market
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1. Introduction

The value of an electricity option is the expectation of its discounted payoff under a pricing measure. Where options are traded, that measure is recovered from the market: the forward curve fixes the drift and implied volatilities fix the dispersion. Most emerging power markets supply neither in full, and the Turkish market is a representative case. Its organised derivative segment, the *Vadeli Elektrik Piyasası* (VEP) operated by EPİAŞ, lists monthly baseload futures, while options on power are not traded. A call written on the day-ahead price of a given delivery hour therefore has to be valued with two gaps: the forward curve is observed only as a strip of monthly averages, and the distribution around that curve is not observed at all.

Each gap is addressed by a mature literature. On the distributional side, regime switching is the established description of power prices. Following Hamilton (1989), spikes have been modelled as visits to a separate state rather than as jumps added to a single diffusion (Deng, 2000; Huisman and Mahieu, 2003), and heavy-tailed regime specifications have been found to fit spot data better than single-regime alternatives (Weron, 2009; Janczura and Weron, 2010). Jump-diffusion and threshold models (Cartea and Figueroa, 2005; Geman and Roncoroni, 2006) and their empirical comparison (Benth et al., 2012) complete the picture, which is surveyed by Weron (2014) and Deschatre et al. (2021). Once transition probabilities are allowed to depend on observed variables (Diebold et al., 1994; Filardo, 1994), load, reserve margin and intra-day conditions are found to carry information on switching (Mount et al., 2006; Kanamura and Ōhashi, 2008; Karakatsani and Bunn, 2008). Residual demand, load net of non-dispatchable renewable output, is the natural covariate (Wagner, 2014; Do et al., 2021), and the merit-order literature explains the mechanism (Sensfuß et al., 2008; Ketterer, 2014; Clò et al., 2015). For Türkiye the merit-order effect has been measured by Sirin and Yilmaz (2020), and the forecasting properties of spot prices have been documented by Ugurlu et al. (2018) and Ertuğrul et al. (2022). Because prices at and below zero have become common, the area hyperbolic sine has largely replaced the logarithm as the transformation of choice (Schneider, 2011; Uniejewski et al., 2018).

On the curve side, the decomposition of a power price into a deterministic seasonal level and a mean-reverting residual (Lucia and Schwartz, 2002),

together with its two-factor commodity counterpart (Schwartz and Smith, 2000), underlies most subsequent work. Building a granular curve from contracts that deliver over a period is posed as a smoothing problem constrained to reproduce the quoted averages (Fleten and Lemming, 2003; Benth et al., 2007), and hourly versions have been constructed for European markets (Caldana et al., 2017; Kiesel et al., 2019). Forward dynamics have been modelled semi-parametrically (Borak and Weron, 2008) and with factor models that respect delivery-period averaging (Kiesel et al., 2009; Benth et al., 2008b).

Valuation on a regime-switching state is equally well developed. The Markov-modulated framework goes back to Naik (1993) and was extended by Buffington and Elliott (2002) and Jobert and Rogers (2006), while changes of measure acting on the switching intensities are treated by Elliott et al. (2005). Finite-difference schemes for the coupled systems are given by Boyle and Draviam (2007) and Huang et al. (2011), applied to gas storage by Chen and Forsyth (2010), and compared with Monte Carlo by Zeng et al. (2017). For electricity, derivatives are priced within a regime-switching model through explicit risk premia by Janczura (2014), and spot and forward information are combined under regime shifts by Paraschiv et al. (2015), the closest antecedent to the present study.

What remains open is a question of identification. When a market supplies a forward strip and nothing else, it has to be established which parameters of such a model are determined by data and which are assumptions brought in by the analyst. The distinction is easily obscured. A model that reproduces the strip to numerical precision appears calibrated, yet volatilities and switching parameters estimated under the physical measure $\mathbb{P}$ and used under $\mathbb{Q}$ without adjustment carry the assumption that no premium is charged for volatility or regime risk. Forward premia in power markets are documented empirically (Longstaff and Wang, 2004; Viehmann, 2011; Weron and Zator, 2014; Algieri et al., 2021), explained in equilibrium (Bessembinder and Lemmon, 2002; Benth et al., 2008a) and linked to systematic forecast errors (Redl et al., 2009). A premium that has not been estimated is not thereby equal to zero.

Contribution.. A valuation framework is built here around that separation, and four contributions follow from it.

First, a forward-centred representation is proposed in which the spot price is written as $P_t = F(t) + X_t - \mathbb{E}^{\mathbb{Q}}[X_t]$, with F an equality-constrained hourly forward curve and X a two-regime Ornstein–Uhlenbeck residual with time-varying transition probabilities. Subtracting the residual’s own mean

makes $\mathbb{E}^{\mathbb{Q}}[P_t] = F(t)$ an identity for every admissible parameter vector (Proposition 1). The identity is a design property rather than a deep result, but it insulates the one moment identified by the market from those that are not, and it continues to hold under the two risk-premium channels implemented here, a shift in the residual drift and a change in the switching intensities.

Second, the inherited residual parameters are tested rather than assumed. The residual process is re-estimated by maximum likelihood on a series constructed to match the model, net of the calendar-month level and the hour-of-week shape. The mean-reversion speed and the innovation scale of the production parameters each differ markedly from these estimates, yet the two discrepancies offset: the implied stationary standard deviation of the residual is 582.5 TRY/MWh at the valuation forward, against 531 to 624 TRY/MWh realised in 2025. Because option values at the reported maturities of 24 to 72 hours are governed by that variance, the 72-hour call at a strike of 3 000 TRY/MWh remains between 169 and 176 TRY/MWh across the estimated range of mean-reversion speeds when the variance is held fixed, whereas values at maturities below about 12 hours are not reliable. Time-varying transitions are strongly supported by likelihood-ratio tests.

Third, the framework is evaluated out of sample. Under a valuation rule that respects day-ahead publication times, predictive distributions are formed on 60 dates in 2026 with every parameter frozen at the end of 2025. The error is dominated by the forward curve, which exceeds realised prices by 385 to 794 TRY/MWh on average across horizons, while the realised residual is 1.33 to 1.55 times wider than the model implies. When the variance comparison is repeated at earlier dates, agreement holds from mid-2023 to the end of 2025, with ratios of realised to implied dispersion between 0.93 and 1.13, and fails in 2026. The failure is traced to the level-proportional scaling implied by a fixed-scale transformation, which is identified as the main specification limit of the model.

Fourth, the monthly strip is shown to bound what forward-anchored valuation can deliver in this market. When the curve construction is replayed at seven valuation dates from 2022 onwards, the December 2025 strip, above which realised monthly prices fell by 1 019 TRY/MWh on average, is found to lie inside a distribution whose bias is large and unstable in sign. The daily reference price of the monthly contracts changed on 1.0% of contract-days for 2026 deliveries, so a delta hedge in the strip had no effectiveness over the evaluation window. To the authors' knowledge, the predictive and hedging

performance of the Turkish monthly strip has not previously been measured.

The market, contract and data are described in Section 2, the hourly curve in Section 3, the residual dynamics and the centring identity in Section 4, and the valuation equations in Section 5. Estimation and validation of the residual process are reported in Section 6, results in Section 7, and their implications and limits in Section 8; Section 9 concludes.

2. Market, contract and data

Wholesale electricity in Türkiye is cleared by EPIAŞ. The day-ahead market sets an hourly price, the *Piyasa Takas Fiyatı* (PTF), for each hour of the following day, and the auction results for day $D+1$ are published at about 14:00 local time (TRT, UTC+3) on day D (EPIAŞ, 2025). Prices are quoted in Turkish lira per megawatt-hour and are bounded by administratively set floor and ceiling levels (Enerji Piyasası Düzenleme Kurumu, 2024). The forward segment, VEP, lists monthly baseload futures that settle against the arithmetic mean of PTF over all hours of the delivery month. A quote therefore restricts the average of 672 to 744 hourly forward prices and is silent on how that average is distributed across hours, which is the first source of the identification problem studied here.

2.1. The contract

The instrument valued is a European call on the PTF of a single expiry hour T ,

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(P_T - K)^+], \quad (1)$$

with strike K and a flat continuously compounded rate r . The payoff is written on one hourly price and not on the monthly average that VEP settles, and the two are kept apart in the implementation: a monthly quote cannot be read as a point forward.

2.2. Valuation instant and samples

Because day-ahead prices are published a day in advance, the valuation instant has to be placed so that no payoff is already known. Valuation is set at 11:00 TRT on 31 December 2025. At that time the last published price is that of 31 December, 23:00 TRT, which equals 2,917.78 TRY/MWh and serves as the observed spot, and all maturities are counted from that hour. The forward curve is built from the last VEP reference prices published

before 11:00, those of 30 December. They are identical to the quotes of 31 December for every contract, so the curve, and every option value reported below, coincides with a construction that uses the year-end quotes.

Six monthly contracts were quoted, for delivery from February to July 2026 (Table 1); January carried no quote. The strip falls from February to June and then rises by 58% into July, a seasonal step that the hourly curve has to accommodate.

Table 1: VEP monthly baseload reference prices used for the valuation of 31 December 2025. January 2026 was not quoted.

Contract	Delivery month	Hours	Price (TRY/MWh)
EBM0226	February 2026	672	2,900.99
EBM0326	March 2026	744	2,556.31
EBM0426	April 2026	720	2,500.66
EBM0526	May 2026	744	2,506.25
EBM0626	June 2026	720	2,245.33
EBM0726	July 2026	744	3,558.19

Three further samples are used. Hourly PTF from January 2019 to the valuation hour (61,368 observations) is used to re-estimate the residual process in Section 6. The transition covariate is hourly residual demand, load net of wind and solar output, standardised with the scaler of the original estimation sample. Realised PTF from 1 January to 27 September 2026 serves only for evaluation. No observation after the valuation cutoff enters any calibration, estimation or model choice, and this is enforced by tests that fail if a fitting routine is given a later timestamp.

3. The hourly forward curve

The framework is organised so that market information and historical estimates enter through separate paths that meet only at the centring step (Figure 1). This section describes the market path.

3.1. Averaging constraint and construction

Let $\mathcal{H}_m$ be the delivery hours of month m and VEP_m its quote. Settlement against the monthly mean requires

$$\frac{1}{|\mathcal{H}_m|} \sum_{h \in \mathcal{H}_m} F(h) = \text{VEP}_m, \quad m \in \mathcal{M}, \quad (2)$$

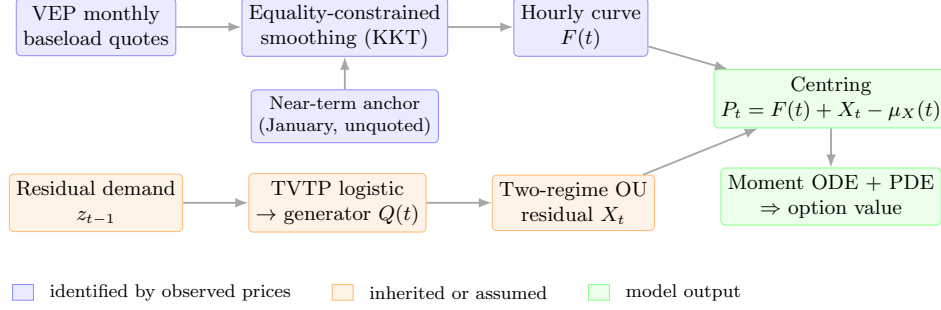


Figure 1: Structure of the framework. Elements in blue are fixed by observed prices and elements in orange are inherited from historical estimation or assumed. The two paths meet only in the centring step, so that the forward level is the only part of the price distribution determined by the market.

which fixes six linear functionals of a curve with $N = 5,089$ hourly values. The remaining degrees of freedom are chosen by a smoothness criterion, which is an assumption and is reported as one. Following Fleten and Lemming (2003) and Benth et al. (2007), the curve $\mathbf{F} \in \mathbb{R}^N$ solves

$$\min_{\mathbf{F}} w_R \|D_2 \mathbf{F}\|_2^2 + w_A \|\mathbf{F} - \mathbf{B}\|_2^2 \quad \text{subject to} \quad A\mathbf{F} = \mathbf{b}, \quad (3)$$

where D_2 is the second-difference operator, $\mathbf{B}$ a baseline equal to VEP_m within quoted months and to the anchor rule of Section 3.2 elsewhere, and each row of A is one instance of (2) or the anchor condition. The problem is solved as one sparse Karush–Kuhn–Tucker system,

$$\begin{bmatrix} 2(w_R D_2^\top D_2 + w_A I) & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} \mathbf{F} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} 2w_A \mathbf{B} \\ \mathbf{b} \end{bmatrix}. \quad (4)$$

Because the quotes enter as equality constraints, the weights shape the curve within months but cannot trade fit against smoothness. The monthly reproduction error, 3.6×10^{-12} TRY/MWh, is therefore a property of the solver rather than evidence for the curve. Hourly increments average 0.45 TRY/MWh in absolute value, and the three largest lie at month boundaries where the strip itself steps, the largest being +47.0 TRY/MWh between June and July (Figure 2a).

3.2. The unquoted front month

The front month is never quoted. At each of the seven valuation dates examined in Section 7.5, from December 2022 to December 2025, the contract

for the next delivery month had already been delisted, so a short-dated option always falls in a window without a forward price. The level there has to be assumed. Three rules were compared: holding the spot flat, holding the next quote flat, and interpolating linearly from the spot at the last known hour to the first quote at the month boundary. The linear rule is used because it satisfies $F(t_0) = P_{t_0}$, joins the February quote with a step of 0.05 TRY/MWh against 12 to 47 TRY/MWh at quoted boundaries, and does not assert that January is a high-price month. Holding the February quote flat violates the spot condition by 17 TRY/MWh. How much the choice matters is shown in Figure 2b and Section 7.2: a shift of $\pm 20\%$ in the anchored level moves the 72-hour call at $K = 3,000$ from 17.80 to 528.96 TRY/MWh.

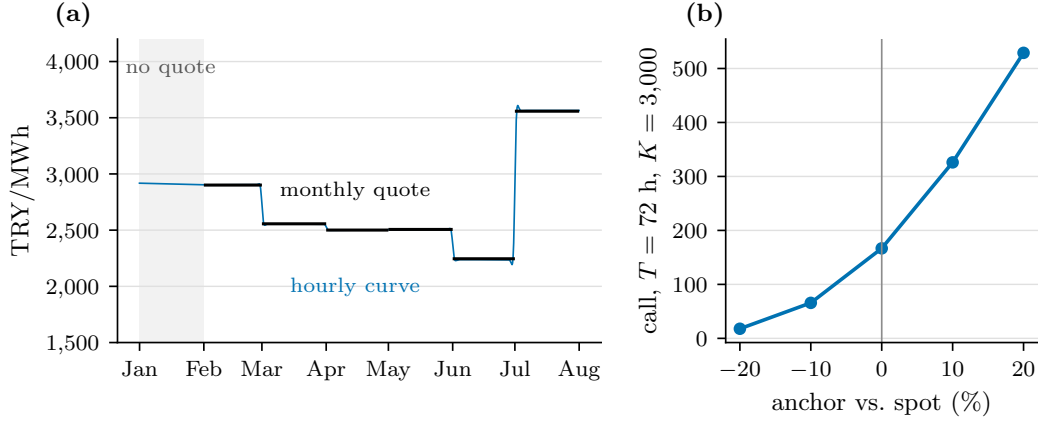


Figure 2: The forward level of the front month is assumed, and it dominates the option value. (a) Hourly forward curve and the six monthly quotes; the shaded January window carries no quote. (b) Value of the 72-hour call at $K = 3,000$ TRY/MWh when the anchored January level is shifted relative to the spot, all other inputs fixed.

4. Residual dynamics and the centring identity

4.1. Transform and regimes

Prices are bounded below by an administrative floor, can be zero or negative in other markets (Aust and Horsch, 2020), and are heavy-tailed above. The estimation underlying the production parameters works with the area hyperbolic sine (Schneider, 2011; Uniejewski et al., 2018),

$$y_t = \operatorname{asinh}(P_t/s_P), \quad P_t = s_P \sinh(y_t), \quad (5)$$

with scale $s_P = 282.48$ TRY/MWh. A continuous-time Markov chain $S_t \in \{0, 1\}$ distinguishes a calm regime 0 and a stress regime 1 with transform-scale volatilities $\sigma_0^y = 0.003535$ and $\sigma_1^y = 0.092407$ per $\sqrt{\text{hour}}$. The labels refer to variance states of the residual, not to economic stress.

4.2. Time-varying transition probabilities

Hourly transition probabilities depend on standardised residual demand at a one-hour lag,

$$p_{01}(t) = \Lambda(\alpha_{01} + \gamma_{01}z_{t-1}), \quad p_{10}(t) = \Lambda(\alpha_{10} + \gamma_{10}z_{t-1}), \quad (6)$$

with $\Lambda(x) = 1/(1 + e^{-x})$, following Diebold et al. (1994) and Filardo (1994). Consistency of maximum likelihood for this class is established by Pouzo et al. (2022), and a score-driven alternative is given by Bazzi et al. (2017). Because z_{t-1} is observed before the transition it governs, the specification is predictive. Over most of the covariate range the calm-to-stress probability exceeds the reverse one, and the two cross near $z = 1.4$ (Figure 3a). The one-step matrix is mapped to a generator by the matrix logarithm,

$$Q(t) = \begin{bmatrix} -q_{01} & q_{01} \\ q_{10} & -q_{10} \end{bmatrix}, \quad q_{ij}(t) = \frac{\lambda(t) p_{ij}(t)}{s(t)}, \quad \lambda(t) = -\ln(1 - s(t)), \quad (7)$$

with $s = p_{01} + p_{10}$; the embedding is exact whenever $s < 1$, which holds throughout the sample. The regime-label reconciliation required by the conversion is described in Appendix B.

4.3. Residual process

Conditional on the regime, the residual follows an Ornstein–Uhlenbeck process in price space,

$$dX_t = [\kappa(m_{S_t} - X_t) + a_{S_t}] dt + \sigma_{S_t}^X(t) dW_t^{\mathbb{Q}}, \quad (8)$$

with a common mean-reversion rate κ , regime means m_i and a drift shift a_i that is zero except where a risk premium is studied (Section 5.3). The diffusion is mapped from the transform scale by the delta method evaluated at the forward level,

$$\sigma_i^X(t) = \sigma_i^y \sqrt{F(t)^2 + s_P^2}, \quad (9)$$

so that price-scale dispersion is proportional to the price level whenever $F \gg s_P$. This assumption is tested in Section 7.4. The process is additive

and can in principle produce negative prices, and a shared κ lets regimes enter the second moment but not the speed of reversion, which keeps the moment system linear. With the production parameters the residual standard deviation saturates within about a day (Figure 3b).

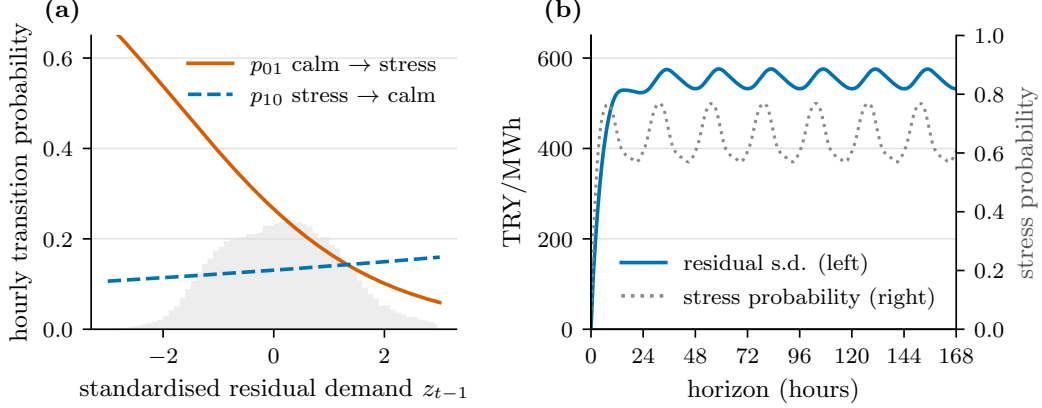


Figure 3: Residual demand moves the chain between variance regimes, and the residual reaches its stationary dispersion within a day. (a) Hourly transition probabilities against lagged standardised residual demand, with the sample distribution of the covariate in grey. (b) Residual standard deviation and stress-regime probability along the pricing covariate path.

4.4. Forward centring

The spot price is defined as

$$P_t = F(t) + X_t - \mu_X(t), \quad \mu_X(t) = \mathbb{E}^{\mathbb{Q}}[X_t]. \quad (10)$$

Proposition 1 (First-moment anchoring). *Let X follow (8) with the regime chain generated by (7), let $\mu_X(t) = \mathbb{E}^{\mathbb{Q}}[X_t]$ be computed under the same measure and parameters, and let P be defined by (10). Then $\mathbb{E}^{\mathbb{Q}}[P_t] = F(t)$ for all t and for every admissible parameter vector*

$$(\kappa, \sigma_0^y, \sigma_1^y, m_0, m_1, a_0, a_1, \alpha_{01}, \alpha_{10}, \gamma_{01}, \gamma_{10}, \pi_0),$$

including any change of the transition intensities of the form $q_{ij}^{\mathbb{Q}} = q_{ij} e^{\eta_{ij}}$.

Proof. By linearity, $\mathbb{E}^{\mathbb{Q}}[P_t] = F(t) + \mathbb{E}^{\mathbb{Q}}[X_t] - \mu_X(t) = F(t)$. The substance lies in the requirement that μ_X be the same functional of the parameters

that drives the dynamics. Section 5.1 gives $\mu_X = u_0 + u_1$, where u_i solve the moment system (12), which is integrated jointly with the regime probabilities used in pricing; the cancellation therefore holds to integration tolerance for any generator, including a transformed one. $\square$

The identity confines every error in the residual layer to the shape of the distribution around a forward level that stays fixed by the quotes. An earlier multiplicative specification, $P_t = s_P \sinh(Y_t)$ with Y a regime-switching process, lacks this property: with the same parameters its implied expected spot converges to about $-3,108$ TRY/MWh against a forward of $+2,916$ TRY/MWh, because the regime-weighted mean of Y is negative. Forward consistency is either imposed or not obtained.

5. Valuation

5.1. Moment system

The centring term and the terminal distribution both follow from regime-partitioned moments,

$$p_i(t) = \mathbb{Q}(S_t = i), \quad u_i(t) = \mathbb{E}^\mathbb{Q}[X_t \mathbf{1}_{\{S_t=i\}}], \quad w_i(t) = \mathbb{E}^\mathbb{Q}[X_t^2 \mathbf{1}_{\{S_t=i\}}], \quad (11)$$

which satisfy the closed linear system

$$\begin{aligned} \dot{p}_i &= -q_{ij}p_i + q_{ji}p_j, \\ \dot{u}_i &= \kappa(m_i p_i - u_i) + a_i p_i - q_{ij}u_i + q_{ji}u_j, \\ \dot{w}_i &= 2\kappa(m_i u_i - w_i) + 2a_i u_i + (\sigma_i^X)^2 p_i - q_{ij}w_i + q_{ji}w_j, \end{aligned} \quad (12)$$

for $i \in \{0, 1\}$ and $j \neq i$, started from $p_i(0) = \pi_i$ and $u_i(0) = w_i(0) = 0$, where π is the filtered regime distribution at the last known hour. The structure is recursive, p being autonomous, u depending on p and w on both, so no closure approximation is needed. Then $\mu_X = u_0 + u_1$ and $\text{Var}^\mathbb{Q}[X_t] = w_0 + w_1 - \mu_X^2$; the derivation is given in Appendix A. Because q_{ij} and σ_i^X vary along the covariate path, the system is non-autonomous and is integrated with sub-hourly steps (Appendix C).

5.2. Pricing equation and verification

Conditional on the regime, the value function $V_i(x, t)$ solves the coupled system

$$\partial_t V_i + [\kappa(m_i - x) + a_i] \partial_x V_i + \frac{1}{2} (\sigma_i^X(t))^2 \partial_{xx} V_i - r V_i + q_{ij}(t) (V_j - V_i) = 0, \quad (13)$$

for $j \neq i$, with terminal condition $V_i(x, T) = (F(T) + x - \mu_X(T) - K)^+$, in the form of Buffington and Elliott (2002) and Jobert and Rogers (2006). It is discretised by Crank–Nicolson finite differences (Boyle and Draviam, 2007; Huang et al., 2011) on 1,201 nodes, and the option value is the π -weighted mixture of $V_0(0, 0)$ and $V_1(0, 0)$. The moment system and the pricing equation draw a_i and q_{ij} from one parameter field, since a perturbation applied to only one of them breaks Proposition 1 without any visible symptom in prices; a regression test requires $\max_t |\mathbb{E}^\mathbb{Q}[P_t] - F(t)| < 10^{-8}$ TRY/MWh under asymmetric perturbations.

Put–call parity holds to 1.0×10^{-6} TRY/MWh across the strike–maturity grid. A direct simulation of (8) with exact conditional Ornstein–Uhlenbeck steps on the sampled regime path gives, with 5×10^5 paths, 166.11 ± 0.40 TRY/MWh for the 72-hour call at $K = 3,000$, against 166.71 TRY/MWh from the finite-difference solution on 2,401 nodes, inside the 95% interval. The production grid value is 166.75 TRY/MWh, and Richardson extrapolation places the discretisation error at 0.04% (Appendix C).

5.3. Risk-premium channels

The residual parameters are estimates under $\mathbb{P}$, so using them for valuation asserts that neither volatility nor regime risk is priced. The strip cannot identify either premium, and both are implemented as explicit channels whose default reproduces the zero-premium values exactly. The first is the drift shift a_i in (8), in the spirit of Janczura (2014). It enters the first moment through $a_i p_i$ and the second through $2a_i u_i$, so its effect on the variance, and through it on option value, is of second order: a log-log regression of the price change on a_1 has slope 2.0000. The second channel acts on the switching intensities,

$$q_{ij}^\mathbb{Q}(t) = q_{ij}(t) e^{\eta_{ij}}, \quad (14)$$

a change of measure of the type studied by Elliott et al. (2005). The multiplicative form keeps the generator valid for every real η , and the same transformed generator enters (12) and (13), so Proposition 1 continues to hold. Because η moves the stationary regime occupancy, its effect on the terminal variance is of first order.

The size of either premium cannot be estimated without option prices. An ex-post premium panel, built from seven VEP snapshots and realised prices up to the valuation cutoff, has five or six observations per horizon, a pooled mean of about 625 TRY/MWh and block-bootstrap standard errors of 280 to

530 TRY/MWh. Translated through $|a| = \kappa L$, the pooled mean corresponds to $|a| \approx 49$ TRY/MWh per hour, which is used below as an upper bound for the drift channel. No comparable anchor exists for η , and the range $|\eta_{ij}| \leq 0.5$ is a scenario rather than an estimate.

6. Parameter provenance and independent estimation

6.1. Provenance

Each parameter is classified as *calibrated* (fixed by market prices), *inherited* (taken from a historical estimation under $\mathbb{P}$ without adjustment), *derived* (reconstructed from summary statistics of that estimation, without a standard error) or *assumed*. Only the forward curve is calibrated (Table 2). The transition intercepts were not reported by the original estimation and were reconstructed by matching the reported mean regime durations of 3.76 and 7.56 hours (Appendix B). The scale s_P is documented as a training median of absolute prices, but the window that produced 282.48 TRY/MWh is not available: the same definition gives 1,399.99 TRY/MWh on 2019–2025 and 302.02 TRY/MWh on 2019–2020, so the value is consistent with an early, pre-depreciation window.

Table 2: Production parameters, their provenance, and maximum-likelihood estimates on the model-consistent residual over 2019–2025 (standard errors in parentheses). Only $F(\cdot)$ is identified by market prices.

Parameter	Production	Status	Re-estimated
$F(\cdot)$	exact on six quotes	Calibrated	–
January level	linear anchor	Assumed	–
κ (h^{-1})	0.0784	Inherited	0.152
σ_0^y	0.00354	Inherited	0.078 (0.006)
σ_1^y	0.0924	Inherited	0.320 (0.011)
α_{01}	–1.016	Derived	–2.615 (0.038)
α_{10}	–1.895	Derived	–1.071 (0.038)
γ_{01}	–0.584	Inherited	–0.759 (0.034)
γ_{10}	+0.078	Inherited	+0.692 (0.041)
s_P (TRY/MWh)	282.48	Assumed	–
π_1 at last known hour	0.068	Inherited	–
a_i, η_{ij}	0	Assumed	–
r (annual)	0.40	Assumed	–

6.2. *Re-estimation on a model-consistent residual*

The inherited parameters were tested by estimating the residual process independently. The object that (8) describes is a deviation from the forward curve, so the estimation series is built to match it: the transformed price minus its calendar-month mean minus the within-month hour-of-week shape. The two-regime model with time-varying transitions is estimated by maximum likelihood with the filter of Hamilton (1989), using multiple starting points and Hessian-based standard errors (Appendix D).

Three results stand out. First, mean reversion on this residual is fast: single-regime fits give κ between 0.19 and 0.26 h^{-1} in every year from 2019 to 2025, and the two-regime fit gives $\kappa = 0.152 \text{ h}^{-1}$ with an interior autoregressive coefficient of 0.859. The production value of 0.078 h^{-1} is reached in no year. Second, the production innovation scale is too narrow, 222 against 359 TRY/MWh per hour in 2025. Third, time-varying transitions are strongly supported: the likelihood-ratio statistic against constant transitions is 1,851 on two degrees of freedom.

The origin of the inherited persistence is visible from a fit to the transformed price level itself, without the monthly level removed. That fit places the autoregressive coefficient on the unit-root boundary, as the inherited estimate does, because over 2019–2025 the level is dominated by lira depreciation. A two-by-two comparison of scale and level removal confirms the attribution: removing the calendar-month level changes the estimated κ by a factor of six to ten, whereas changing the scale from the transform to lira changes it by a factor of 1.3 to 1.7. An earlier version of the present model inherited $\kappa = 4.11 \times 10^{-6} \text{ h}^{-1}$, a half-life of nineteen years, from such a fit and priced the reference call at 687.04 TRY/MWh, four times its present value. Persistence of the transformed price level and persistence of deviations around a forward curve are different objects, and only the second belongs in (8).

6.3. *What the inherited parameters get right*

Mean reversion and innovation scale are individually wrong in opposite directions, and the errors offset in the quantity that governs option values at the reported maturities, the stationary variance. A regime-mixture approximation gives a stationary residual standard deviation of 582.5 TRY/MWh at the valuation forward, and along the covariate path used in pricing the terminal standard deviation at 72 hours is 532.3 TRY/MWh. The realised value for 2025 is 531 TRY/MWh when the hour-of-week shape is estimated on 2025 alone and 624 TRY/MWh when it is pooled over 2019–2025. In a

long-path simulation the Kolmogorov–Smirnov distance to the realised 2025 residual is 0.087 for the production parameters, against 0.125 for the full-window re-estimate and 0.159 for a fit restricted to 2022–2025; no parameter set is accepted by the test, partly because the realised series is truncated by the price limits. The re-estimated sets imply stationary deviations of 929 and 1,088 TRY/MWh and overstate 2025 dispersion. The production parameters are therefore retained for pricing, and the re-estimated full-window set is carried as an alternative in the out-of-sample evaluation.

Holding the stationary variance at its production value isolates the role of κ (Figure 5a). In a single-regime reduction, the 72-hour call at $K = 3,000$ stays between 169 and 176 TRY/MWh across the estimated range $\kappa \in [0.15, 0.22] \text{ h}^{-1}$ and between 162 and 187 TRY/MWh across $[0.01, 0.30]$, while the 1-hour call moves from 7 to 98 TRY/MWh. Mean reversion therefore matters at short maturities and not at the 24 to 72 hours reported here, and values below about 12 hours are not reliable under the production parameters. The same reduction also measures what the regime mixture contributes at equal variance: at the production κ the single-regime call is 183.4 TRY/MWh against 166.75 TRY/MWh for the two-regime process, a difference of -9% that reflects the heavier tails of the mixture.

7. Results

7.1. Option values and benchmarks

At the reference contract, a call with $K = 3,000$ TRY/MWh expiring 72 hours after the last known hour, the forward is $F(T) = 2,916.16$ TRY/MWh, the residual standard deviation 532.26 TRY/MWh and the call value 166.75 TRY/MWh. Because the residual saturates within a day, values change little with maturity once the forward is fixed (Table 3), and the simulated probability of a negative terminal price is zero to four decimal places.

Three closed-form benchmarks are priced on the same forward and discount factor: Black-76 and Bachelier with volatilities estimated from daily price changes over 2019–2025, and the Lucia and Schwartz (2002) model with the production κ and the regime volatilities pooled at the stationary occupancy. Near the money the Lucia–Schwartz value lies about 10% above the model, whereas the two models without mean reversion diverge with maturity, reaching +118% (Bachelier) and +195% (Black-76) at 168 hours (Figure 4a). Against payoffs realised in 2026 over a grid of 66 strike–maturity pairs, the

mean absolute errors are 151 TRY/MWh for the model, 157 TRY/MWh for Lucia–Schwartz, 272 TRY/MWh for Bachelier and 369 TRY/MWh for Black-76. Expressed as Black-76 implied volatilities, the model produces a smile that declines in strike at every maturity and a term structure that falls from 3.29 at 24 hours to 0.61 at 720 hours at $K = 3,000$, the signature of a mean-reverting residual (Figure 4b).

Table 3: Call values at the valuation date (TRY/MWh). The last row gives the mean absolute error against payoffs realised in 2026 over 66 strike–maturity pairs.

T (h)	K	$F(T)$	Model	Lucia–Schwartz	Bachelier	Black-76
24	2,000	2,917.24	927.11	927.64	917.28	918.27
24	3,000	2,917.24	163.93	182.45	114.63	163.68
24	4,000	2,917.24	5.59	5.39	0.26	7.60
72	2,000	2,916.16	924.70	925.25	938.28	947.85
72	3,000	2,916.16	166.75	184.07	223.74	308.58
72	4,000	2,916.16	5.94	5.73	14.09	74.40
168	2,000	2,913.99	918.57	919.12	1,007.26	1,029.72
168	3,000	2,913.99	164.95	182.16	359.38	486.06
168	4,000	2,913.99	5.84	5.63	72.63	218.09
MAE against 2026 payoffs			151.37	156.65	272.47	369.19

7.2. Sensitivity to unidentified inputs

Table 4 varies one unidentified input at a time and orders the inputs by their effect on the reference call. The ordering is the main result.

The forward level over the unquoted month dominates every other input by an order of magnitude. Near the money, moneyness drives value, and a $\pm 20\%$ shift moves the forward by about 580 TRY/MWh against a residual dispersion of 530 TRY/MWh. Where the front month is unquoted, the precision of any reported value is therefore bounded by the uncertainty in that level, and no refinement of the residual layer can improve on the bound.

The transition-intensity premium is the second channel. Raising η_{01} and lowering η_{10} shifts occupancy toward the stress regime and raises the call by up to 21.1%; the opposite combination lowers it by 30.3% (Figure 5b), and with the drift premium added at its upper bound the range widens to -38.6% and $+29.1\%$. The drift premium alone reaches $+5.7\%$ at the bound

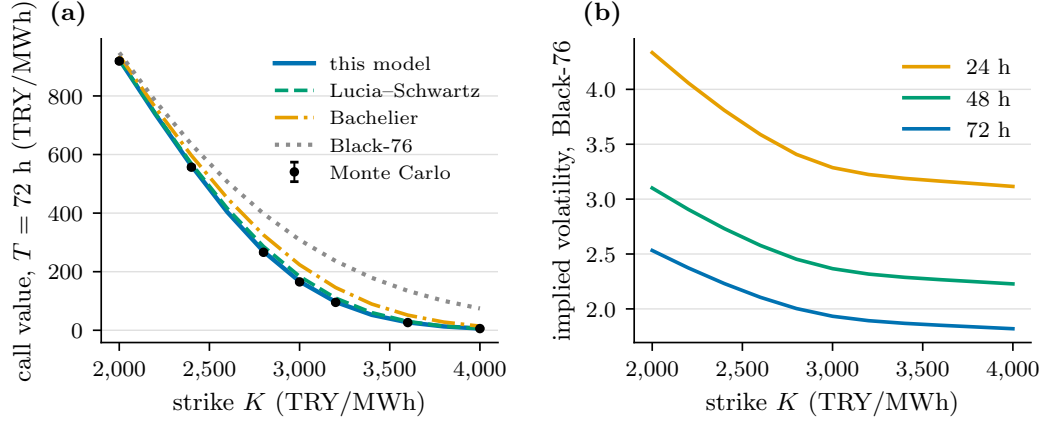


Figure 4: Mean reversion separates the model from benchmarks without it. (a) Call values at 72 hours against strike for the model, three closed-form benchmarks, and Monte Carlo simulation of the model with 95% intervals. (b) Black-76 volatilities implied by model prices at three maturities.

Table 4: Effect of unidentified inputs on the 72-hour call at $K = 3,000$ TRY/MWh (baseline 166.75 TRY/MWh). Each row varies one input with all others at production values.

Input	Variation	Change in value
January forward level	$\pm 20\%$	-89.3% to +217.2%
Intensity premium	$ \eta_{ij} \leq 0.5$	-30.3% to +21.1%
Residual-demand path	± 2 s.d. shift	-35.2% to +20.4%
Regime mixture	one regime, same variance	+10.0%
Drift premium	$a_1 \leq 50$ TRY/MWh/h	+5.7%
Mean reversion	$\kappa \in [0.15, 0.22]$, same variance	spread 3.9%
Ramp covariate	estimated, occupancy fixed	-1.0%
Regime volatilities	alternative model	-0.7%
Discount rate	$r \in [0.15, 0.60]$	-0.2% to +0.2%
Initial regime distribution	stationary, not filtered	< 0.001%

of Section 5.3, consistent with its second-order effect. A premium on regime risk is therefore the one that option data, if they became available, would identify most readily.

The residual-demand channel has a definite sign: shifting the covariate path down by two standard deviations raises the call by 20.4% and shifting it up lowers it by 35.2%, because $\gamma_{01} < 0$ makes high residual demand suppress transitions into the high-variance regime. Low residual demand is the condition in which renewable output displaces the flexible thermal units that would otherwise absorb short-run imbalances, so the direction is plausible, but it is reported as a property of the estimated coefficients. The second covariate of the original estimation, the hourly change of residual demand, alters the call by -1.0% when its estimated slopes are added with the mean regime durations held fixed, so its omission from the production model is immaterial at this maturity.

The remaining inputs are second order at 72 hours. Mean reversion at fixed stationary variance was discussed in Section 6.3. The initial regime distribution is forgotten within a few hours, since the regime memory has a half-life of 1.37 hours: replacing the filtered distribution by one filtered under the re-estimated parameters moves the call by 20.8% at 1 hour, 0.06% at 24 hours and not at all at 72 hours. The irrelevance belongs to the horizon, not to the input. The discount factor at 720 hours moves only from 0.988 to 0.952 across the range considered.

7.3. Out-of-sample evaluation in 2026

Predictive distributions were formed on 60 valuation dates, every third business day from 5 January to 24 September 2026, under the timing rule of Section 2.2, with horizons of 6 to 72 hours after the last known hour. On each date the forward curve was rebuilt from the latest quotes and every parameter was held at its end-2025 value. The production parameters are compared with the full-window re-estimate of Section 6 and with the three benchmarks, using the continuous ranked probability score (CRPS), the probability integral transform (PIT), the tests of Berkowitz (2001) and Diebold and Mariano (1995), and interval coverage (Diebold et al., 1998; Gneiting and Raftery, 2007).

The predictive mean equals the forward, so the error separates into a forward-curve component and a residual component. Using the realised 2026 monthly level and month-specific hourly shape as a reference, the forward curve exceeds realised prices by 385 to 794 TRY/MWh on average, with a

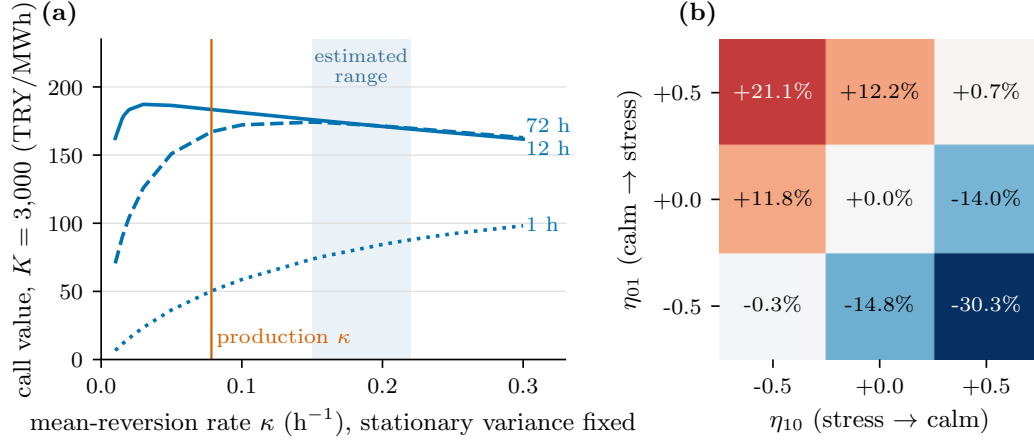


Figure 5: Mean reversion matters at short maturities only, and a premium on regime risk moves values far more than a drift premium. (a) Call at $K = 3,000$ TRY/MWh against κ with the stationary variance held at its production value, single-regime reduction, for maturities of 1, 12 and 72 hours; the shaded band is the range estimated in Section 6. (b) Change in the 72-hour call under the intensity premium $q_{ij}^{\mathbb{Q}} = q_{ij}e^{\eta_{ij}}$.

Table 5: Out-of-sample evaluation over 60 dates in 2026 (TRY/MWh). Forward error is the forward curve minus the realised monthly level and hourly shape; the residual ratio compares realised and model residual standard deviations. p is the Diebold–Mariano p -value for equal CRPS of the two parameter sets.

h	CRPS		p	PIT mean	Forward error		Residual ratio
	Production	Re-estimated			Mean	s.d.	
6	466.8	429.0	0.015	0.33	463	767	1.55
12	781.5	733.6	< 0.001	0.21	794	843	1.33
24	502.5	494.8	0.43	0.40	385	697	1.52
48	730.2	699.7	0.008	0.35	427	716	1.51
72	756.3	727.2	0.014	0.30	484	717	1.35

day-to-day standard deviation of 697 to 843 TRY/MWh (Table 5, Figure 6a). The forward curve carries all of the bias and about half of the error variance. The realised residual is 1.33 to 1.55 times wider than the model implies, part of which reflects the rise in dispersion from 2025 to 2026 documented in Section 7.4. PIT values pile up near zero (Figure 6b), as expected when realised prices fall below a forward that overshoots, and 90% intervals cover between 37% and 68% of outcomes.

The wider re-estimated set scores better on CRPS at four of five horizons, but it prices options worse. Near the money, its mean pricing error against realised payoffs is 101 to 184 TRY/MWh, against 13 to 64 TRY/MWh for the production set, and its tail is too heavy. A good density score and an unbiased option price are different targets here, because the dominant error is in the level and a wider distribution transmits a level error more fully into option value. The production set also outperforms the Lucia–Schwartz benchmark on CRPS at four horizons ($p < 0.001$), whereas Bachelier, whose dispersion grows without bound, scores better at 48 and 72 hours. Across moneyness, in-the-money calls are overpriced and out-of-the-money calls underpriced, the pattern of a forward that is too high combined with a residual that is too narrow.

Delta hedging with the monthly contract was evaluated on the same dates. The sensitivity of the forward at the target hour to the monthly quote rises from 0.09 at 6 hours to 0.35 at 72 hours, but the hedge earned exactly zero because the reference price of the relevant contract did not change within any evaluation window. Over 2022–2026 the daily reference price changes on 8.5% of contract-days; for contracts delivering in 2026 the share is 1.0%, and on quotation days in 2026 it is 0.6%. The monthly strip neither spans hourly risk nor moves often enough to hedge with.

7.4. Stability of the variance agreement

The agreement of Section 6.3 was checked at seven earlier dates and at the end of the evaluation sample. At each date the residual was constructed from the preceding twelve months only, and its standard deviation was compared with the production value implied by (9) at the twelve-month mean price L , which is close to $0.199 L$. From mid-2023 to the end of 2025 the ratio of realised to implied dispersion lies between 0.93 and 1.13, with a mean of 1.02 (Figure 7a). It is 1.26 for 2022, a year of extreme prices, 1.65 for the twelve months to September 2026 and 1.91 for 2026 alone; a pooled hourly shape raises every ratio by 9 to 18%.

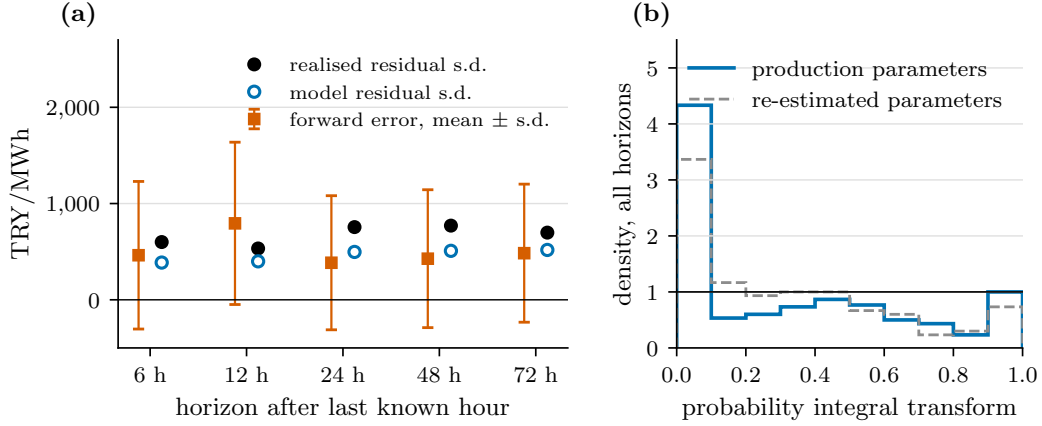


Figure 6: Out-of-sample error in 2026 is dominated by the forward curve. (a) By horizon: mean and standard deviation of the forward-curve error, realised residual standard deviation and model residual standard deviation. (b) Histogram of probability integral transforms pooled over horizons; a uniform histogram would lie on the horizontal line.

The 2026 departure is not a level effect, since the implied dispersion is evaluated at each window’s own mean price. Realised dispersion relative to the price level rises from 0.19–0.23 over 2023–2025 to 0.33 and 0.38 in the two 2026 windows (Figure 7b). Hours below 50 TRY/MWh rise from under 1% to 5–7% of the sample in 2026, but they occur at midday, are largely absorbed by the hourly shape, and excluding them leaves the ratios almost unchanged. The scaling built into a fixed-scale transform with delta mapping, dispersion proportional to the price level, therefore holds over 2023–2025, is exceeded in 2022 and fails in 2026.

7.5. Predictive error of the monthly strip

Because the curve reproduces the quotes exactly, any error in the level is inherited from the strip. Against realised prices from January to July 2026, 5,088 hours, the December 2025 strip exceeded realised monthly means by 1,019 TRY/MWh on average, by 24% in July and 76% in May (Figure 8a). In May, 222 of 744 hours cleared below 10 TRY/MWh. The period coincided with strong hydrological conditions and record renewable output (Sirin and Yilmaz, 2020), and the deviation is described rather than attributed. January, whose level came from the anchor, shows a bias of -14.47 TRY/MWh, which reflects the information in a spot observed one hour before the window opened rather than any merit of the interpolation rule.

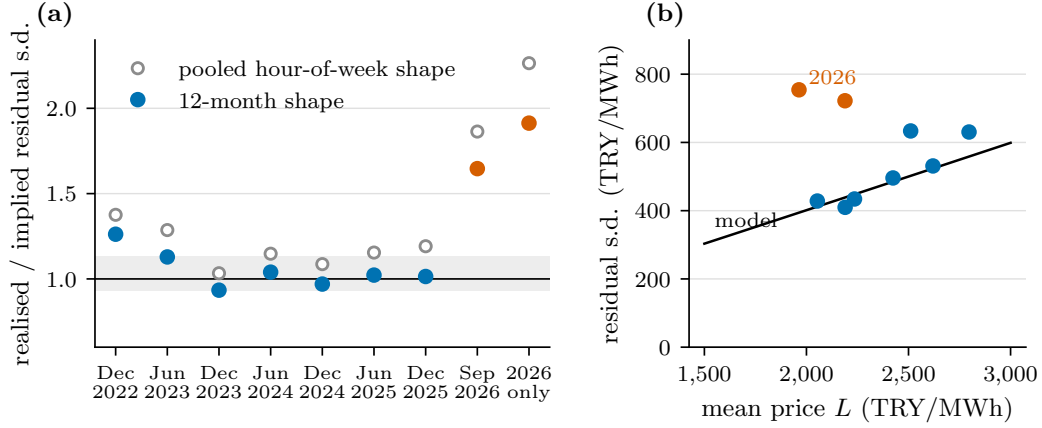


Figure 7: The production variance matches realised dispersion from mid-2023 to 2025 and fails in 2026. (a) Ratio of realised to model-implied residual standard deviation by date, twelve-month window; the band marks 0.93 to 1.13. (b) Realised residual standard deviation against the twelve-month mean price, with the model’s proportional scaling.

The curve construction contains no historical parameter and can be replayed at earlier dates without look-ahead. At seven dates from December 2022 to December 2025, 49 date-month pairs, realised prices fell below the strip at five dates and above it at two, with date means between $-1,465$ and $+290$ TRY/MWh (Figure 8b). The pooled mean error is -603 TRY/MWh, or -18.2% , with a standard deviation of 775 TRY/MWh, and the 2026 outcome lies at the 33rd percentile of the pooled distribution. Aggregated to one observation per date the mean gives $t = -2.29$ ($p = 0.062$), and windows that do not overlap give p between 0.11 and 0.49, so the strip is described as a biased and sign-unstable predictor without a calibrated significance level. Across the seven anchored months the realised relative error of the anchor has a standard deviation of 11.7% , close to the 11.5% of a uniform $\pm 20\%$ band, and a mean of -3.5% ; the band used in Table 4 has the right width but is not a containment bound.

8. Discussion and limitations

8.1. What a monthly strip identifies

Calibration to a baseload strip fixes six linear functionals of the hourly curve and, through Proposition 1, the first moment of the price at every

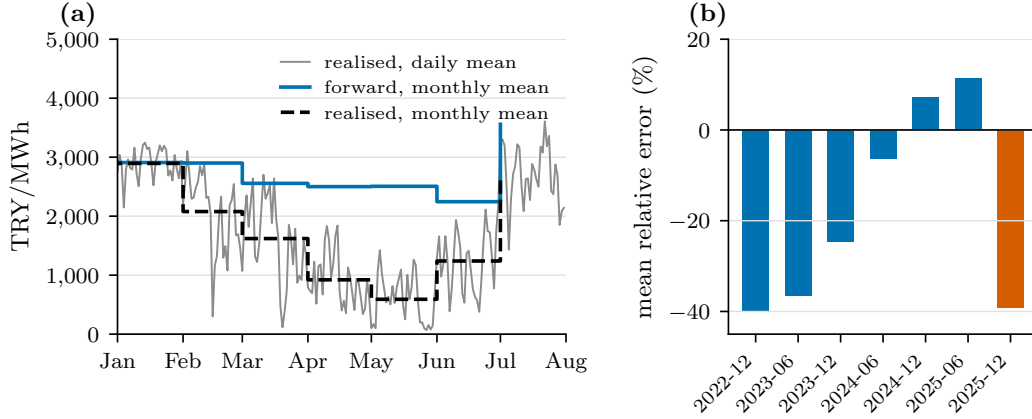


Figure 8: The monthly strip is a biased and sign-unstable predictor of realised prices. (a) Realised prices in 2026 against the forward curve of 31 December 2025. (b) Mean relative error of the strip at seven valuation dates; 31 December 2025 is highlighted.

horizon. It does not fix the intraday shape, the level over the unquoted month, the dispersion around the curve or any premium, and the largest single driver of option value is an input that no observed price constrains. A reporting convention follows. In markets of this kind a value is better stated conditional on an explicit anchor and accompanied by the range that plausible alternatives induce; for the reference call the $\pm 20\%$ band gives 17.80 to 528.96 TRY/MWh around 166.75 TRY/MWh, and the multi-date evidence of Section 7.5 supports that width. The centring identity is what makes this practical, because premia, anchors and residual parameters can be varied without disturbing the calibration.

8.2. What the strip predicts

Whether forward prices are unbiased predictors of later spot prices is an old question in power markets, framed by hedging pressure (Bessembinder and Lemmon, 2002), by risk premia whose sign depends on the horizon (Benth et al., 2008a), and by systematic forecast errors (Redl et al., 2009). On Turkish monthly contracts the strip is neither unbiased nor reliably biased in one direction: the pooled error averages -18% of the forward level, yet two of seven dates reversed the sign. The out-of-sample evaluation adds the hourly view, in which the forward curve carries all of the bias of the predictive distribution. Separating a premium from a forecast error would require either

an instrument for expectations or a structural model of the supply stack, and seven dates cannot identify either. For a book anchored to this curve, position risk is dominated by the level it is anchored to: a standard deviation of 775 TRY/MWh in the level error compares with a residual dispersion of about 530 TRY/MWh at the horizons studied.

8.3. *Specification limits*

The main limit is the scaling of dispersion with the price level. A fixed-scale transform combined with the delta mapping (9) makes the residual standard deviation proportional to the forward, and Section 7.4 shows that this holds over 2023–2025 and fails in 2026, when relative dispersion rose by more than half. A scale that adapts to the price regime, or a residual volatility that depends on the level explicitly, is the natural extension. It cannot be added without care, because the centring identity requires the first moment of the residual to remain computable under the same measure.

The residual is additive and unbounded below while realised prices face administrative limits. A specification with a slow second factor and explicit price limits was developed alongside the present one, but clipping the state distribution breaks the centring identity, so restoring $\mathbb{E}^{\mathbb{Q}}[P_t] = F(t)$ would require a redesigned centring term. The alternative of a multiplicative or logarithmic form respects the lower bound automatically but loses forward consistency through Jensen’s inequality, at a cost measured in Section 4 in thousands of TRY/MWh, whereas the simulated probability of a negative price under the additive form is zero to four decimal places.

Three parameters remain imperfectly documented. The transition intercepts are reconstructions without standard errors, the scale s_P is consistent with an early estimation window that cannot be verified, and the regime labels were reconciled from an output in which the high-variance state is the dominant one. The independent estimation of Section 6 bounds the consequences: replacing all four transition coefficients by maximum-likelihood estimates from the transformed price level changes the reference call by +3.75 TRY/MWh, and the inherited volatilities are retained because they reproduce the realised stationary variance, not because they are individually correct.

8.4. *Scope of the evaluation*

The evaluation covers one market, one year of out-of-sample data and seven forward-curve dates whose delivery windows partly overlap. Four of the seven strips were published one to four days before their valuation date

because the date fell on a non-business day, and a strict same-day rule would leave three dates. No option is traded on the underlying, so option values are evaluated against realised payoffs rather than market prices, and no premium can be calibrated. Extending the panel as quotes and outturns accumulate, and calibrating the intensity premium if option or structured-contract prices become observable, are the natural continuations.

9. Conclusion

Where a market supplies a monthly forward strip and no options, the parts of an option value that are identified and the parts that are assumed can be kept apart by construction. An equality-constrained hourly curve carries the identified level, a regime-switching residual with time-varying transitions carries the distributional assumptions, and a centring term ensures that the second cannot alter the first under any parameter vector or change of measure.

Three implications follow for valuation in the Turkish market and in markets like it. The forward level of the unquoted front month bounds the precision of any short-dated option value, and a value is more informative when reported with the range that this level induces. Residual parameters inherited from another estimation need to be tested on a series that matches their use; here they were individually wrong but correct in the stationary variance that governs the reported maturities, and that agreement held from 2023 to 2025 and failed in 2026, when dispersion stopped scaling with the price level. Finally, forward-anchored valuation inherits the predictive accuracy of the strip, which was low and unstable in sign over 2022–2026, and the strip moved too rarely to hedge with.

A residual whose dispersion adapts to the price regime, and a premium on regime risk calibrated once option-like prices become observable, are the extensions that the evidence points to.

Data and code availability

The code, processed data and output files behind every table and figure are archived at <https://doi.org/10.5281/zenodo.23083820>. Raw EPIAŞ data remain subject to the terms of use of the EPIAŞ transparency platform.

CRediT authorship contribution statement

Onat Uzkan: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft. **Koray Ömercan Sağlı:** Methodology, Software, Validation, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with code development and testing, the verification of numerical results against the output files, and language editing of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Appendix A. Derivation of the moment system

Let $f_i(x, t)$ be the joint density of $(X_t, S_t = i)$ under $\mathbb{Q}$, with X following (8) and the chain generated by $Q(t)$, possibly transformed as in (14). The forward equation for the pair is

$$\partial_t f_i = -\partial_x[(\kappa(m_i - x) + a_i) f_i] + \frac{1}{2} (\sigma_i^X)^2 \partial_{xx} f_i + q_{ji} f_j - q_{ij} f_i, \quad j \neq i. \quad (\text{A.1})$$

Multiplying by 1, x and x^2 and integrating over $\mathbb{R}$, with boundary terms vanishing because f_i and $x f_i$ decay at infinity, gives (12) for $p_i = \int f_i dx$, $u_i = \int x f_i dx$ and $w_i = \int x^2 f_i dx$. Time dependence enters through $q_{ij}(t)$ along the covariate path and through $\sigma_i^X(t)$ via (9).

For a drift shift of size a , write $u_i = u_i^{(0)} + a u_i^{(1)} + O(a^2)$. The u block gains the forcing $a_i p_i$ and is first order in a ; the w block gains $2a_i u_i$, so the change in variance, $w_0 + w_1 - \mu_X^2$, is of order a^2 . This concerns the variance channel. For a contract away from the money the regime reweighting induced by a_i can contribute a first-order term to the option value, and the numerical results of Section 7.2 bound the total effect rather than establishing a general second-order result. A change of intensities, by contrast, enters the p block directly and moves the stationary occupancy, and hence the variance, at first order in η .

Appendix B. Transition reconstruction and label reconciliation

The original estimation output orders regimes so that state 0 carries the higher volatility, occupies about 67% of the sample and has a mean duration of 7.56 hours, while state 1 has a mean duration of 3.76 hours. The convention here is the reverse. Adopting it requires swapping the volatilities and, because the logistic coefficients are indexed by the same labels, swapping the transition coefficients crosswise, so that the production γ_{01} is the original γ_{10} and conversely. Applying the swap to only one of the two objects would leave volatilities and transitions referring to different states without any single diagnostic revealing it; the ordering $\sigma_0^y < \sigma_1^y$ is enforced when the parameters are loaded.

The output supplies slopes and mean durations but not intercepts. The intercepts solve

$$\mathbb{E}_z[\Lambda(\alpha_{01} + \gamma_{01}z)] = \frac{1}{d_{\text{calm}}}, \quad \mathbb{E}_z[\Lambda(\alpha_{10} + \gamma_{10}z)] = \frac{1}{d_{\text{stress}}}, \quad (\text{B.1})$$

with the expectation taken over the full standardised covariate sample and target rates of 0.2660 and 0.1323 per hour. The resulting pair reproduces the reported stationary stress share of 0.6746 to within 1.0% under an ergodic approximation and 4.5% under simulation along the realised covariate path. These values are moment-matching reconstructions without a sampling distribution, which is why they are labelled as derived in Table 2.

Appendix C. Numerical implementation and convergence

The moment system (12) is advanced by fourth-order Runge–Kutta with sub-steps of at most 0.25 hours and coefficients interpolated linearly within

each step; on hourly steps without sub-stepping the second-moment trajectory was unstable at the estimated switching intensities. The pricing equation (13) is solved by Crank–Nicolson on a uniform grid in the residual coordinate whose half-width is six terminal standard deviations, with two time steps per hour and the regime coupling treated in the same implicit step as the diffusion.

Table C.6: Spatial convergence of the 72-hour call at $K = 3,000$ TRY/MWh.

Nodes	Value (TRY/MWh)	Change from previous
301	167.0716	–
601	166.8415	–0.2301
1,201	166.7477	–0.0938
2,401	166.7070	–0.0407
4,801	166.6940	–0.0130
Richardson limit		166.686

Table C.6 reports spatial convergence. The observed order is 1.2 to 1.7 rather than the second order of Crank–Nicolson for smooth data, which is the expected consequence of the kink in the payoff (Rannacher, 1984; Giles and Carter, 2006). Richardson extrapolation from the two finest grids gives 166.686 TRY/MWh, so the production grid of 1,201 nodes carries an error of 0.06 TRY/MWh, or 0.04%. Refining the time step eightfold at 2,401 nodes changes the value by 0.011 TRY/MWh, below the spatial error, and varying the domain half-width between four and nine standard deviations changes it by at most 0.024%. The discretisation error is therefore two orders of magnitude below the effects reported in Table 4. Put–call parity, the centring identity under asymmetric premium perturbations, and agreement with Monte Carlo simulation are run as automated regression tests.

Appendix D. Estimation details

Residual construction. Let $y_t = \text{asinh}(P_t/s_P)$ for hourly PTF from January 2019 to the valuation hour. The model-consistent residual is $y_t - \bar{y}_{m(t)} - g_{m(t)}(h(t))$, where $\bar{y}_m$ is the mean of y over calendar month m and g_m the mean deviation from it at hour-of-week h within that month. The construction removes the monthly level, which the forward curve supplies in the pricing

model, and the hourly shape, which the curve also carries, so that what remains corresponds to X in (8).

Likelihood.. The two-regime autoregression with regime-dependent mean and volatility and logistic transitions (6) is estimated by maximising the likelihood computed with the filter of Hamilton (1989). Each fit uses twenty random starting points and a bounded quasi-Newton optimiser; regime labels are ordered by volatility after estimation, and standard errors are taken from the inverse Hessian, with outer-product estimates as a check. The mean-reversion rate is $\kappa = -\ln \phi$ per hour. Likelihood-ratio statistics for time-varying against constant transitions are referred to a χ^2_2 distribution.

Boundary estimates.. On the transformed price level without the monthly level removed, the autoregressive coefficient converges to the unit-root boundary. Standard asymptotics do not apply there (Andrews, 2001), and the fit is reported through the profile likelihood over a grid of fixed ϕ , whose maximum lies at 0.99999 with a positive definite Hessian for the remaining parameters. These boundary fits are used only to explain the origin of the inherited persistence and do not enter pricing.

Real prices.. Deflating PTF by the consumer price index of TÜİK (series TP.GENENDEKS.T1, base 2003 = 100) to December 2025 prices gives a scale of 3,092.05 TRY/MWh under the documented definition and regime volatilities 22 to 33% lower than in nominal terms, so inflation accounts for part, but not all, of the gap between inherited and re-estimated volatilities.

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